

MELBOURNE AP, AustraliaWMO: **948660**Lat: **37.6656S**Lon: **144.8322E**Elev: **119**StdP: **99.91**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.8	3.9	-2.8	3.0	21.7	-1.0	3.5	16.3	16.0	11.2	14.5	11.1	3.6	0	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	35.4	18.1	32.9	17.8	30.2	17.4	20.8	28.1	19.8	27.5	18.9	26.4	7.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	22.7	17.7	12.8	21.6	16.7	12.1	20.9	60.7	27.8	57.2	27.7	54.1	26.5	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.3	10.8	DB	0.4	40.6	0.8	2.4	-0.1	42.3	-0.6	43.7	-1.1	45.0	-1.6	46.8	(4)
(5)			WB	0.0	22.9	0.8	1.4	-0.5	23.9	-1.0	24.7	-1.4	25.5	-2.0	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	20.2	20.3	18.3	15.1	12.4	10.1	9.5	10.2	11.9	14.0	16.3	18.1	(6)	
	DBStd	4.94	4.13	3.92	3.65	3.06	2.45	1.94	1.81	2.16	2.71	3.41	3.86	4.04	(7)	
	HDD10.0	92	0	0	0	1	4	22	31	24	8	2	0	0	(8)	
	HDD18.3	1635	23	21	47	106	185	247	274	253	194	144	87	54	(9)	
	CDD10.0	1792	318	287	257	153	78	24	16	29	66	125	188	252	(10)	
	CDD18.3	293	82	75	45	9	1	0	0	0	1	8	25	47	(11)	
	CDH23.3	3428	952	789	469	85	2	0	0	0	11	132	345	644	(12)	
(13)	CDH26.7	1549	488	370	186	15	0	0	0	0	1	38	136	314	(13)	
(14)	Wind	WSAvg	5.3	5.3	5.0	4.9	4.7	5.0	5.2	5.8	5.8	5.5	5.2	5.3	(14)	
(15) Precipitation	PrecAvg	537	43	44	34	46	41	39	36	46	48	55	58	47	(15)	
	PrecMax	821	102	201	86	142	156	105	94	97	127	144	158	139	(16)	
	PrecMin	310	4	1	4	5	8	12	7	16	11	16	18	2	(17)	
	PrecStd	121	24	49	19	33	29	21	20	21	26	32	36	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	37.8	35.7	28.7	22.7	17.6	17.0	20.1	24.7	30.8	33.9	38.0	(19)	
		MCWB	19.1	19.1	17.9	15.2	13.4	12.3	11.2	12.8	14.4	15.4	17.0	18.1	(20)	
	2%	DB	35.6	34.1	31.2	25.8	20.0	16.1	15.1	17.6	21.3	27.0	30.7	34.0	(21)	
		MCWB	18.4	18.3	17.3	14.9	13.2	11.8	10.5	11.4	13.1	14.5	16.6	17.2	(22)	
	5%	DB	32.1	31.2	28.2	23.3	18.2	14.9	14.0	15.9	19.2	23.9	27.2	30.1	(23)	
		MCWB	18.2	18.2	16.9	14.6	12.7	10.9	9.8	10.5	12.4	13.9	16.1	16.7	(24)	
	10%	DB	28.5	28.2	25.5	21.2	16.9	13.9	13.0	14.3	17.4	21.1	24.2	26.6	(25)	
		MCWB	17.8	18.0	16.5	14.1	12.1	10.4	9.4	9.7	11.5	12.9	15.4	16.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	22.1	20.4	17.6	15.8	14.0	12.5	13.5	15.9	17.6	20.1	20.8	(27)	
		MCDB	29.1	28.9	28.1	24.4	18.9	15.9	15.4	18.1	20.6	24.9	27.6	27.4	(28)	
	2%	WB	20.9	20.7	19.2	16.5	14.3	12.2	11.3	12.1	14.2	16.1	18.5	19.1	(29)	
		MCDB	28.2	28.0	26.0	21.8	18.0	14.8	14.0	16.7	19.3	22.7	25.6	27.0	(30)	
	5%	WB	19.8	19.7	18.3	15.6	13.4	11.5	10.5	11.0	13.1	14.9	17.3	18.1	(31)	
		MCDB	28.4	26.8	24.6	20.6	16.9	14.1	13.0	14.7	18.2	21.5	24.3	26.3	(32)	
	10%	WB	18.6	18.7	17.3	14.7	12.6	10.7	9.9	10.2	11.9	13.7	16.1	17.1	(33)	
		MCDB	26.8	25.8	23.5	19.8	16.1	13.3	12.2	13.5	16.6	19.8	22.6	24.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	11.7	10.9	10.0	7.9	7.0	6.9	8.1	9.6	11.1	11.5	12.1	(35)	
		MCDBR	19.4	17.6	16.2	13.7	9.8	8.5	8.1	10.1	12.5	16.3	17.6	19.5	(36)	
		MCWBR	5.8	5.7	5.8	5.9	5.1	5.0	4.7	5.4	6.2	6.8	6.6	6.3	(37)	
	5% WB	MCDBR	14.8	14.1	12.8	11.3	8.4	6.9	6.9	9.2	11.2	13.4	14.0	14.9	(38)	
		MCWBR	5.7	5.6	5.5	5.5	4.9	4.5	4.5	5.3	6.1	6.5	6.3	6.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.347	0.351	0.326	0.326	0.309	0.298	0.298	0.311	0.329	0.333	0.345	0.339	(40)	
	taud		2.508	2.504	2.574	2.540	2.573	2.592	2.584	2.531	2.480	2.488	2.480	2.520	(41)	
	Ebn at Noon		986	958	944	879	837	819	842	883	925	965	981	999	(42)	
	Edh at Noon		112	108	93	84	70	64	68	83	100	108	114	112	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.18	6.27	5.01	3.48	2.30	1.78	1.99	2.81	4.01	5.35	6.29	7.06	(44)	
	RadStd		0.46	0.33	0.29	0.21	0.14	0.14	0.12	0.19	0.29	0.43	0.39	0.40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.47	N/A	N/A	+0.83	N/A	N/A	-19	-117	+159	+58
(47)	Regional (38 neighbors)	+0.51	N/A	N/A	+0.88	N/A	N/A	-17	-125	+174	+60

Nomenclature: See separate page